PHYSICAL CLIMATE RISK: FLOOD MODEL COMPARISON, HAZARD AND VULNERABILTY ASSESSMENT AND FINANCIAL IMPLICATIONS

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Master Thesis

NASLOV V SLOVENŠČINI

Magistrsko delo

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Abstract

Flooding is one of the most frequent and damaging natural hazards globally, and its risk is projected to increase due to climate change. Accurate flood risk assessments are essential for spatial planning, climate adaptation, and ESG-aligned financial decision-making. This thesis investigates physical flood risk by comparing global and local hazard models, evaluating standardized vulnerability functions, and quantifying financial implications for real estate assets in Slovenia.

Two flood hazard models are assessed: the global WRI Aqueduct River Flood model and the high-resolution local model Integralna karta globin (IKG). A geospatial validation dataset comprising flood depth and damage records from four flood events in Slovenia (2010–2023) is used to evaluate model accuracy. Flood depths are analyzed using both regression and classification metrics, while vulnerability is assessed through comparison of observed damages against standardized depth–damage functions provided by the European Commission Joint Research Centre (JRC).

A key technical contribution is the development of a data processing pipeline that harmonizes heterogeneous IKG hazard data and integrates it into the modular Physrisk platform using the Zarr data format. This enables scalable, scenario-based risk assessment and model intercomparison.

Findings suggest that while local models offer improved accuracy for continuous flood depth prediction in regression tasks, the global model performs better in categorical classification. Proximity to water bodies was not found to significantly improve prediction accuracy. Depth–damage functions align well with observed damage distributions for residential properties, though segmentation by property type reveals variability in model performance.

The thesis contributes to the validation and integration of physical climate risk models for real estate applications and supports the advancement of ESG-compliant climate risk disclosures.

Povzetek

Povzetek v slovenščini naj ne bo daljši od ene strani.

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Abbreviations

JSI ... Jožef Stefan Institute
IPS ... International Postgraduate School
AEP ... Average Exceedance Probability
EAD ... Expected Annual Damage
ESG ... Environmental, Social, and Governance
JRC ... Joint Research Centre
RP ... Return Period
WRI ... World Resources Institute

Symbols

$j^\star$ ... black-body irradiance
σ ... Stefan's (or Stefan-Boltzmann) constant

Glossary

Glossary of terms

Chapter 1

Introduction

1.1 Background and Motivation

Flooding is a major natural hazard, responsible for substantial economic losses, social disruption, and loss of life across the globe. With increasing urbanization and changing climate conditions, flood risks are intensifying in both frequency and magnitude. The Intergovernmental Panel on Climate Change (IPCC) projects a rising likelihood of extreme hydrological events, including both pluvial and fluvial floods, due to higher precipitation variability and rising sea levels.

In Europe, and particularly in Central and Eastern regions such as Slovenia, recent flood events have exposed vulnerabilities in both infrastructure and institutional response systems. The need for robust flood risk models that can inform spatial planning, insurance pricing, and asset management is therefore more pressing than ever. This is especially relevant for stakeholders in the real estate and financial sectors, who must now account for physical climate risks as part of their environmental, social, and governance (ESG) reporting and compliance requirements.

1.2 Climate Risk and the ESG Framework

Climate-related physical risks, such as floods, heatwaves, and storms, are becoming a critical component of ESG disclosures. Investors and financial institutions are increasingly required to assess and report the exposure of their assets to climate hazards. In this context, the concept of physical risk is typically decomposed into three components:

- **Hazard** – the probability and intensity of the climate-related event (e.g., flood depth, wind speed).
- **Vulnerability** – the expected damage to an asset given the intensity of the hazard.
- **Exposure** – the value or presence of assets in harm's way.

The combination of these three factors forms the basis of quantitative risk models that estimate potential economic losses due to climate events. Yet, despite the wide adoption of such frameworks, the underlying models are often poorly validated, especially at local or parcel-level scales relevant for property-level decision making.

1.3 Problem Statement

While global flood hazard models such as the WRI Aqueduct River Floods provide broad coverage and climate scenario integration, they are often limited in spatial resolution and may not reflect local hydrological conditions. On the other hand, high-resolution local models, such as Slovenia's *Integralna karta globin* (IKG), offer detailed inundation mapping but are typically fragmented, static (i.e., not climate scenario-aware), and difficult to integrate into broader risk platforms.

Furthermore, flood vulnerability is typically modeled using generalized depth–damage functions derived from empirical or synthetic data. These functions may perform well in aggregate, but their applicability across different building types, geographies, and damage-reporting standards remains uncertain. This creates uncertainty in downstream financial estimates and limits the usefulness of such models for asset-level ESG disclosure.

1.4 Research Objectives

This thesis aims to evaluate the performance of global and local flood risk models with a specific focus on hazard and vulnerability components. The main objectives of the research are:

1. To compare the predictive accuracy of flood depth estimates from a global model (WRI Aqueduct) and a high-resolution Slovenian model (IKG) using real flood observations.

2. To assess how well global depth–damage functions (e.g., from the JRC) reflect reported flood damage across different real estate asset types in Slovenia.

3. To evaluate whether spatial resolution or proximity to water bodies significantly influence model performance.

4. To develop and document a data onboarding pipeline that enables integration of heterogeneous local flood maps into the modular *Physrisk* physical risk modeling framework.

1.5 Methodological Approach

The thesis adopts a comparative modeling approach using geospatial and statistical techniques. A validation dataset, compiled from the records of the Slovenian Administration for Civil Protection and Disaster Relief, includes appraised damage and measured flood depths for selected events between 2010 and 2023.

Flood hazard models are evaluated using both regression-based and classification-based metrics to assess depth prediction accuracy. Vulnerability models are tested by comparing observed versus predicted damages using segmented analysis by flood depth and real estate type. All data are processed and harmonized into a common geospatial reference system and stored in Zarr format, facilitating efficient access within the *Physrisk* platform.

1.6 Scientific and Practical Relevance

From a scientific perspective, this work contributes to the ongoing effort of validating global hazard and vulnerability models in local contexts. By empirically assessing model

accuracy, the thesis provides critical insight into where global models fall short and where local models offer meaningful advantages.

Practically, the work supports the advancement of data-driven ESG disclosures by enhancing the interoperability and transparency of physical risk assessment tools. The technical contributions also provide a template for integrating fragmented local hazard datasets into scalable, modular frameworks like *Physrisk*, which are increasingly used in institutional risk modeling and regulatory reporting.

1.7 Structure of the Thesis

The remainder of the thesis is structured as follows:

- **Chapter 2** presents a review of relevant literature and related work on flood hazard modeling, vulnerability estimation, and ESG-aligned risk frameworks.
- **Chapter 3** describes the study area, datasets, and the methodological framework used for model evaluation and data processing.
- **Chapter 4** details the integration of the local IKG model into the Physrisk framework and presents the architecture of the comparison toolset.
- **Chapter 5** provides experimental results, including quantitative evaluation of model performance, depth–damage function analysis, and financial impact estimates.
- **Chapter 6** discusses the key findings, limitations, and possible future research directions, and concludes the thesis.

Chapter 2

Floating Bodies

Floating bodies are figures, tables and algorithms.

2.1 Figures

Captions should be placed below figures as shown in Figure 2.1. If a caption is shorter than the line width, it should be centered.

Figure 2.1: A large IPS logo.

On the other hand, if a caption is very long (see Figure 2.2), only its first (short) part should be put in the List of Figures.

Figure 2.2: A small IPS logo. The IPS has its own logo and a uniform graphic image, which is used on all its documents.

2.2 Tables

Similar rules apply also to captions of tables, with the exception that captions are placed above tables (see Table 2.1).

Table 2.1: A simple table.

A	B	C
12	9834	327
51	2234	97

2.3 Algorithms

Algorithm 2.1 presents an algorithm example.

Algorithm 2.1: An algorithm example.

```
Data: this text
Result: complete understanding
initialization;
while not at end of this document do
    read current section;
    if understood then
        go to next section;
        current section becomes this one;
    else
        go back to the beginning of current section;
    end
end
```

Chapter 3

Equations and Measurement Units

3.1 Equations

Small equations are often written in-line (within the text), for example $j^\star = \sigma T^4$, while larger ones need to be displayed in the following way:

$$\sigma = \frac{2\pi^5 k^4}{15c^2h^3} = 5.6704 \times 10^{-8} Js^{-1}m^{-2}K^{-4} \tag{3.1}$$

All displayed equations need to be numbered so that they can be referenced later in the text (for example, Eq. (3.1) presents the Stefan's (or Stefan-Boltzmann) constant).

3.2 Measurement Units

The candidate can choose a standard for measurement units and has to consistently use it throughout the thesis.

Chapter 4

Model Integration and Analysis Framework

Accurate flood risk assessment requires the integration of diverse data sources and computational tools to simulate hazard intensity, estimate potential damages, and evaluate financial exposure. In this chapter, we present the technical framework developed to combine heterogeneous flood hazard data—particularly local Slovenian models—into a unified system suitable for analysis and comparison within the Physrisk platform.

The focus lies on harmonizing the *Integralna karta globin* (IKG) flood maps, which vary in structure, format, and coordinate reference systems, and transforming them into a consistent, machine-readable representation. These models are rasterized, standardized by return period, and encoded in the cloud-optimized Zarr format. The processed datasets are then onboarded into the modular Physrisk platform, which separates hazard, vulnerability, and financial components and allows for flexible model evaluation.

To facilitate empirical comparison, the framework also includes a toolset that enables side-by-side querying and batch evaluation of both global and local models at precise geolocations, aligned with the validation dataset. This infrastructure supports all subsequent experimental analysis presented in Chapter 5 and lays the groundwork for scalable, reproducible climate risk assessments.

The chapter is organized as follows:

- Section 4.1 presents the overall system architecture and the interaction between components.
- Section 4.2 describes the pipeline for transforming and combining IKG models across multiple Slovenian regions.
- Section 4.3 explains how the standardized hazard data are integrated into the Physrisk platform.
- Section 4.4 outlines the model comparison toolset used to evaluate prediction accuracy across models.

4.1 Architectural Diagram

The integration of global and local flood hazard models into a unified analysis platform requires a modular, extensible architecture that supports heterogeneous input formats, scenario-aware modeling, and scalable evaluation. This section presents the architecture developed for this purpose, highlighting its components, data flow, and interoperability within the *Physrisk* framework.

4.1.1 System Overview

Figure **??** illustrates the architecture of the flood hazard model integration pipeline, based on the open-source *OS-Climate Physrisk* ecosystem. The process begins with raw flood hazard data, which may originate from global sources such as the WRI Aqueduct or TU Delft flood models, or from local sources such as the Integralna karta globin (IKG) used in Slovenia. These datasets are onboarded into the `os-risk-hazard` module, a Python-based processing library designed to standardize, harmonize, and prepare hazard indicators for risk evaluation.

The onboarded models are then encoded in Zarr format and stored in an AWS S3-compatible object storage. The Physrisk framework retrieves hazard indicators from this backend and combines them with vulnerability and exposure components to compute impact distributions. The resulting outputs can be accessed via the Physrisk API or used directly for ESG reporting and financial risk analysis through the `os-risk-financial` module.

4.1.2 Data Flow and Model Interoperability

Each flood model—whether global or local—is processed through the same onboarding interface, allowing standardization of outputs while preserving return-period-specific intensity information. The hazard models are structured into three-dimensional Zarr arrays indexed by latitude, longitude, and return period. This enables consistent querying and evaluation across models with different spatial resolutions or geographic coverage.

Once onboarded, all hazard layers become interchangeable within the Physrisk pipeline. Vulnerability models (e.g., depth–damage functions) and exposure datasets (e.g., building values or asset locations) can then be used in combination with any hazard source. This interoperability facilitates both comparative experiments and compound risk evaluations, where multiple models are used to bound uncertainty or stress-test assumptions.

4.1.3 Integration of the IKG Model

The locally developed IKG model, originally provided in shapefile and GeoPackage vector formats, required a dedicated pre-processing stage to align with the Physrisk architecture. This process involved:

- Transforming coordinate systems (from EPSG:3794 to EPSG:4326),
- Rasterizing vector layers into a fixed-resolution 5x5m grid,
- Assigning discrete flood depth values to each return period (RP10, RP100, RP500),
- Converting the raster layers into Zarr format for chunked cloud storage.

The unified model was then uploaded to AWS S3 and integrated into the Physrisk hazard module alongside WRI and TU Delft models. This enabled direct comparison and batch evaluation of global versus local model predictions, described further in Section 4.4.

4.1.4 Hazard Onboarding Pipeline

The transformation from raw IKG data into a cloud-optimized flood hazard model is summarized in Figure **??**. The pipeline consists of the following steps:

1. **Rasterization:** Converting input vector data into gridded raster tiles using a 5x5m resolution grid.

2. **Zarrify:** Chunking the raster into 1000x1000 tiles and encoding into Zarr format.

3. **Upload:** Transferring the data to AWS S3 for use within Physrisk.

The final onboarded dataset can be queried via the Physrisk API or accessed directly for integration with ESG tools or batch model evaluation.

4.2 Pipeline for Combining Heterogeneous IKG Models

The *Integralna karta globin* (IKG) flood hazard datasets are developed for individual municipalities or regions in Slovenia, each based on local hydrological and hydraulic analyses. These datasets differ in file structure, geospatial reference systems, data quality, and encoding of flood depths. To integrate these into a unified flood hazard model suitable for comparative evaluation and platform onboarding, a standardized processing pipeline was developed.

4.2.1 Data Sources and Heterogeneity

The raw IKG datasets were obtained from multiple engineering consultancies (*Inženiring za vode d.o.o., Urbania d.o.o., iS Projekt d.o.o., IZVOR d.o.o., DHD d.o.o.*), each responsible for flood modeling in a subset of areas classified as "Območja pomembnega vpliva" (Areas of Significant Flood Risk) by Slovenian water authorities.

Each local model typically includes:

- GIS vector data (shapefiles, GeoPackages) representing inundation zones,
- Attribute data encoding flood extent and classes for selected return periods (e.g., 10-, 100-, and 500-year events),
- Region-specific coordinate systems (typically EPSG:3794, Slovenia National Grid).

In total, data from 18 areas of importance were collected, with each area providing flood scenarios for three return periods. This resulted in 54 distinct layers that needed to be harmonized.

4.2.2 Standardization and Processing Workflow

To unify the IKG datasets into a single model, the following pipeline was developed and implemented in Python using open-source geospatial processing libraries (GDAL, Fiona, Rasterio, xarray):

1. **Coordinate Transformation:** All source files were transformed from EPSG:3794 to WGS84 (EPSG:4326) to ensure compatibility with global models and standardized grid alignment.

2. **Rasterization:** Vector layers were rasterized into a regular grid with a resolution of 5 meters (5x5m), which approximates the native resolution of the original hydrological models. A fixed flood depth value was assigned to each flood class:

    - GM (minor flooding): 0.5 m
    - GS (significant flooding): 1.5 m
    - GB (severe flooding): 2.5 m

3. **Return Period Alignment:** Each rasterized layer was tagged with its associated return period (RP10, RP100, RP500). For each area, three separate rasters were produced and stacked by return period.

4. **Tiling and Merging:** The processed rasters from all areas were mosaicked into a common spatial grid and stored as a 3D array indexed by latitude, longitude, and return period. Conflicts in overlapping regions (e.g., model boundaries) were resolved by assigning priority to higher-confidence datasets based on metadata quality.

4.2.3 Unified Flood Hazard Product

The final result of this pipeline is a unified flood hazard model for Slovenia covering all 18 areas of importance. This model:

- Captures flood depth estimates at high spatial resolution for three return periods,
- Standardizes input data from diverse local models into a single harmonized format,
- Enables consistent comparison with global models such as WRI Aqueduct within the same framework.

Figure **??** provides a schematic representation of the transformation process, from raw shapefiles to a standardized hazard product.

This unified model was subsequently converted to Zarr format and onboarded into the Physrisk platform, as described in Section 4.3.

4.3 Physrisk Framework – Model Onboarding

Once the local IKG flood models were unified into a standardized and spatially consistent format, the next step involved integrating them into the modular *Physrisk* framework. Physrisk is part of the broader OS-Climate open-source ecosystem and is designed to facilitate scalable, scenario-based physical climate risk analysis. The framework separates hazard, vulnerability, and financial components and enables flexible model composition, comparison, and evaluation.

4.3.1 Physrisk Architecture and Modules

The Physrisk architecture is composed of several interoperable modules:

- **OS-risk-hazard**: responsible for ingesting and formatting hazard data sets (e.g., flood depth by return period) into a common format.
- **OS-risk-physrisk**: computes impact distributions by convolving hazard data with vulnerability functions and exposure information.
- **OS-risk-financial**: translates physical impacts into financial metrics such as loss of asset value or disruption to operations.
- **Physrisk API**: exposes model outputs through a standardized programmatic interface.

The onboarding of hazard models occurs within the `os-risk-hazard` component and requires the hazard data to conform to a Zarr-based storage structure optimized for parallel access and spatial querying.

4.3.2 Zarr Format and Indexing

The Zarr format is a cloud-native format for storing large multi-dimensional arrays. It was chosen for hazard model storage due to its ability to:

- Efficiently compress and store gridded data in chunks (e.g., 1000×1000 tiles),
- Allow selective download of spatial subsets,
- Support multi-indexing (e.g., latitude, longitude, return period).

For the unified IKG model, the hazard data were encoded into a 3-dimensional Zarr array with the following dimensions:

- i: Latitude (y-axis),
- j: Longitude (x-axis),
- k: Return period index, corresponding to RP10, RP100, and RP500.

Each tile was stored as a compressed chunk and uploaded to an AWS S3-compatible bucket accessible to all Physrisk modules.

4.3.3 Onboarding Procedure

The onboarding process was implemented using the Physrisk hazard onboarding tool, a Python-based utility designed for structured hazard model integration. The steps were as follows:

1. **Prepare Metadata:** Define the return periods, CRS (EPSG:4326), spatial resolution, and transformation matrix for the IKG model.
2. **Convert to Zarr:** Use the onboarding tool to encode the processed GeoTIFF files into a Zarr store, assigning appropriate chunk sizes (e.g., 1000×1000 pixels).
3. **Upload to Cloud Storage:** Transfer the Zarr directory to a cloud-accessible S3 bucket used by the OS-Climate infrastructure.
4. **Register the Model:** Add a model reference entry into the hazard configuration of Physrisk, enabling downstream access via API and batch processors.

4.3.4 Integration with Other Models

With the IKG model onboarded, it became available in the same infrastructure alongside global flood hazard layers, including:

- WRI Aqueduct River Flood (global, 1 km resolution),
- TU Delft River Flood (Pan-European, 100 m resolution).

This interoperability allows any of the onboarded hazard layers to be used with common vulnerability functions and exposure datasets, enabling direct comparisons, uncertainty bounding, and ensemble evaluations. The IKG model is now fully accessible via the Physrisk API and can be used for property-level flood risk analysis in Slovenia under a unified modeling interface.

4.4 Comparison Toolset

To evaluate and compare the predictive performance of flood hazard models, a model-agnostic comparison toolset was developed and integrated into the Physrisk framework. This toolset enables side-by-side querying of multiple hazard sources at specific geolocations and return periods, aligned with the reference flood observations used for validation in Chapter 5.

4.4.1 Purpose and Design Principles

The comparison toolset was designed with the following goals in mind:

- **Model-agnostic access:** Allow querying of different onboarded hazard models (e.g., IKG, WRI Aqueduct) through a unified interface.
- **Return-period specificity:** Ensure comparability by extracting flood depth predictions at the same return period (e.g., 100-year flood).
- **Location-accurate evaluation:** Retrieve model outputs at high spatial precision to match reported flood locations.
- **Batch processing support:** Enable evaluation across hundreds of observations using efficient geospatial access patterns.

4.4.2 Interface and Functionality

The toolset consists of a lightweight Python interface built on top of the Physrisk hazard API. Given a set of coordinates (latitude, longitude) and a desired return period, the tool performs the following steps:

1. **Coordinate Transformation:** Convert input coordinates into the grid index of the onboarded Zarr array using affine transformations defined in the model metadata.
2. **Chunk Identification and Retrieval:** Locate the Zarr chunk containing the requested point and load the corresponding depth value for the specified return period.
3. **Output Formatting:** Return the flood depth value, return period, and model metadata for downstream use in regression or classification analysis.

When applied in batch mode, the tool leverages parallel reads and lazy loading to evaluate large datasets (e.g., the full validation set of flood reports) with minimal memory overhead.

4.4.3 Supported Models and Inputs

The toolset was validated for use with the following onboarded hazard models:

- **SI IKG (Unified Model)**: High-resolution local flood model rasterized at 5x5 m, covering 18 Slovenian areas of importance, with RP10, RP100, and RP500 layers.
- **WRI Aqueduct Flood Hazard**: Global model with 1 km resolution, scenario-aware, supporting multiple return periods from RP2 to RP1000.

The input to the toolset is a tabular dataset containing:

- Latitude and longitude of each observation,
- Desired return period (e.g., RP100),
- Optional fields: timestamp, municipality, reported depth, reported damage.

4.4.4 Integration with Validation Workflow

The comparison outputs are stored in a structured table, allowing for direct computation of model performance metrics such as:

- **Regression error:** difference between predicted and observed flood depths,
- **Classification accuracy:** match rate within flood depth categories (e.g., <0.5 m, 0.5–1.5 m, >1.5 m),
- **Bias analysis:** mean and median error segmented by proximity to water, terrain type, or municipality.

These outputs form the quantitative basis for the experimental results presented in Chapter 5.

4.4.5 Reproducibility and Extensibility

The toolset was developed as a standalone module and version-controlled as part of the project's repository. This design ensures that future updates to hazard models (e.g., inclusion of climate scenario variants or additional regions) can be incorporated without changes to the validation logic.

Furthermore, the modularity of the toolset supports extension to:

- Additional hazard types (e.g., pluvial flooding),
- Vulnerability assessments (via model chaining),
- Scenario-based risk projections (by switching hazard layers).

Thus, the comparison toolset serves not only as a core utility for this thesis, but also as a scalable evaluation framework for broader applications in physical climate risk modeling.

Chapter 5

Reference Formatting

References should be formatted using either the IEEE or APA 6th edition formatting style. This template uses the former one.

5.1 IEEE

Let us cite a journal article [2]. References to journal articles that have not yet been published should contain the `doi`, as in [3]. Multiple references can be cited at the same time [4], [5], [6], [7], [8]. Beside books and journal articles, parts of books [9], technical reports [10], PhD theses [11] and MSc theses [12] can be included in the references.

Finally, on-line sources can be referenced, too, see Section **??**.

5.2 APA 6th Edition

Please see the template using the APA style.

Chapter 6

Conclusions

We came to the following conclusions ...

Appendix A

Proofs of Theorems

A.1 Proof of the Pythagorean Theorem

Let us prove the Pythagorean Theorem from page **??**.

Proof. This proof is based on the proportionality of the sides of two similar triangles, that is, upon the fact that the ratio of any two corresponding sides of similar triangles is the same regardless of the size of the triangles.

Let ABC represent a right triangle, with the right angle located at C, as shown in Figure A.1. We draw the altitude from point C, and call H its intersection with the hypotenuse AB. Point H divides the length of the hypotenuse into two parts.

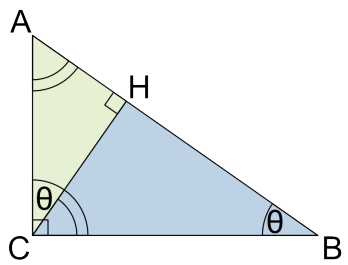

Figure A.1: Similar triangles used in the proof of the Pythagorean theorem.

The new triangle ACH is similar to triangle ABC, because they both have a right angle (by definition of the altitude), and they share the angle at A, meaning that the third angle will be the same in both triangles as well, marked as θ in Figure A.1. By a similar reasoning, the triangle CBH is also similar to ABC.

Similarity of the triangles leads to the equality of ratios of corresponding sides:

$$\frac{BC}{AB} = \frac{BH}{BC} \text{ and } \frac{AC}{AB} = \frac{AH}{AC}. \tag{A.1}$$

The first result equates $\cos\theta$ and the second result equates $\sin\theta$.

These ratios can be written as:

$$BC^2 = AB \times BH \text{ and } AC^2 = AB \times AH. \tag{A.2}$$

Summing these two equalities, we obtain:

$$BC^2 + AC^2 = AB \times BH + AB \times AH = AB \times (AH + BH) = AB^2, \tag{A.3}$$

which, tidying up, is the Pythagorean theorem:

$$BC^2 + AC^2 = AB^2. \tag{A.4}$$

□

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